

Class Notes: Phylum Chordata — General Features and Subphyla

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Defining Chordate Features

Fundamental characters of Chordata: presence of notochord, dorsal hollow nerve cord, and paired pharyngeal gill slits. Bilaterally symmetrical, triploblastic, coelomate, organ system level; possess a post-anal tail and a closed circulatory system.

Chordates and Non-chordates

Phylum Chordata is divided into three subphyla: Urochordata/Tunicata (e.g. Ascidia, Salpa, Doliolum, Herdmania), Cephalochordata (e.g. Branchiostoma/Amphioxus or Lancelet), and Vertebrata/Craniata.

The Three Subphyla

"All vertebrates are chordates, but all chordates are not vertebrates" — in Vertebrata, the embryonic notochord is replaced by a cartilaginous or bony vertebral column in adults. Vertebrata is divided into Agnatha (lacks jaw — Class Cyclostomata) and Gnathostomata (bears jaw), which is further split into superclass Pisces (bear fins — Chondrichthyes, Osteichthyes) and superclass Tetrapoda (bear limbs — Amphibia, Reptilia, Aves, Mammalia).